Spatial–Spectral Feature-Enhanced Mamba and SAM-Guided Hyperspectral Multiclass Change Detection

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***Abstract*— Multiclass change detection from hyperspectral image (HSI) leverages the rich spectral information of HSIs to detect and classify subtle changes in interest in an imaged scene. However, challenges arise due to limited samples in small categories, which hinder the accurate differentiation of changes. This study proposes a spatial–spectral feature-enhanced Mamba and SAM-guided (SFMS) hyperspectral multiclass change detection method. To address the challenges, a tri-plane gated Mamba (tri-Mamba) is designed to complement spatial information using the abundant spectral information in HSIs. In addition, frequency-domain features are combined with state-space models, enabling the detection of more accurate semantic and texture changes using integrated information from frequency domains. This approach effectively mitigates the problem of inaccurate detection in small-sample categories. Furthermore, the segment anything model (SAM) is adapted, with the features of change areas being enhanced through prior knowledge obtained from segmentation, thereby improving the multiclass change detection accuracy. The experimental results demonstrate that the proposed SFMS method outperforms state-of-the-art techniques, achieving superior multiclass change detection while overcoming the challenges associated with detecting small-sample categories.**

***Index Terms*— Mamba, multiclass change detection, segment anything, wavelet.**

I. Introduction

CHANGE detection has long been a significant area of research in computer vision, which focuses on identifying alterations in the same region over different time phases. With the rapid development of remote sensing technology, hyperspectral image (HSI) has gained increasing prominence due to its high spectral resolution. Key study areas related to HSI include target detection [1], [2], [3], classification [4], [5], [6], image fusion [7], [8], [9], and change detection [10], [11], [12]. Among these, HSI change detection remains among the most widely studied and impactful topics. These methods are commonly used to monitor the evolution of natural environmental features [13], [14], as well as for disaster assessment and early warning systems [15]. HSI change detection methods can be binary and multiclass. While binary change detection provides useful information, it is often insufficient for specialized tasks. In contrast, multiclass change detection allows identifying a broader range of changes, through using richer and more detailed information.

HSI change detection can be achieved using three main approaches: traditional algebra-based methods, classical machine learning techniques, and deep learning methods. Early change detection methods primarily relied on algebraic computations, such as change vector analysis (CVA) [16], which identifies changes by calculating the spectral vector difference between images from two phases. However, this method requires selecting an appropriate threshold and often involves complex image preprocessing steps. While traditional methods offer advantages in spectral information processing, they are highly sensitive to environmental variations and noise, resulting in poor performance in complex or dynamic scenes.

Classical machine learning approaches can address the limitations of traditional methods. These augmented methods handle the challenges posed by environmental noise and variations more effectively due to their ability to automate and systematize feature extraction. For example, support vector machines (SVMs) [17] have been widely used in hyperspectral change detection due to their strong generalization capability in high-dimensional spaces and their effectiveness in separating changed and unchanged classes. SVMs construct optimal hyperplanes to distinguish between classes, even in cases where the spectral distributions are complex and nonlinear. In the context of HSI, where redundant and noisy spectral bands are common, SVM-based methods have shown superior performance in maintaining detection accuracy under varying imaging conditions.

Despite these advances, classical machine learning techniques often require manual feature extraction, which, while capable of improving detection performance, depends heavily

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on the quality of the chosen features. The rise of deep learning methods has significantly advanced the accuracy and overall performance of change detection. Unlike traditional methods, deep learning techniques can automatically extract high-level features using models, such as multilayer neural network architectures, making them particularly effective for change detection tasks. Deep learning not only excels at processing large-scale datasets but also captures intricate patterns and has proven robust across various complex scenarios. As a result, applying deep learning in change detection has markedly improved model robustness and precision.

Deep learning offers a range of automatic feature learning techniques for HSI processing, including methods based on convolutional neural networks (CNNs) [18], [19], [20], [21], [22] and transformer-based approaches [23], [24], [25], [26], [27]. Among these, Roy et al. [28] proposed a network that integrates 3-D and 2-D convolutions to extract spectral and spatial features, respectively, where the 3-D convolution first processes the spectral dimension of the image to capture spectral features, while the 2-D convolution operates on the feature map generated by the 3-D convolution to extract spatial features. This dual-convolution approach effectively integrates the information in spectral and spatial dimensions, improving feature extraction performance. Ge et al. [29] introduced a multidirectional and multiscale residual structure, which connects multiple 3-D convolutions of different scales in series to extract spectral features. In addition, 3-D convolutions are applied in multiple directions to capture more comprehensive spatial features. By incorporating multiscale methods, this approach enhances the extraction of contextual information, improving the robustness of feature representation.

Although these methods have shown promising results, CNNs possess inherent limitations, particularly the inability to model long-range dependencies between distant pixels due to their limited receptive fields. In contrast, transformer-based models leverage the self-attention mechanism to effectively addressing the shortcomings of CNNs in capturing global dependencies. Song et al. [30] proposed a cross-temporal attention structure that incorporates the attention mechanism to highlight the differences between two-phase images. This design enhances the focus of the model on the changes between the two images, improving the accuracy of change detection. Chen et al. [31] introduced a three-path local and hybrid-head transformer global network, where a transformer simultaneously extracts local spatial, spectral, and spatial–spectral features, and the improved transformer results are then used to extract global change information. Zhao et al. [32] proposed a graph-based transformer method for HSI change detection, which adopts the transformer architecture as a graph structure to capture long-range change information more effectively. These transformer-based approaches, which aggregate spectral and spatial information, have significantly enhanced the robustness and accuracy of change detection tasks.

However, the self-attention block in transformer models often comes with high computational overhead. In comparison, Mamba [33], [34], which is based on the space-state model (SSM), presents a promising alternative. By leveraging state transitions, Mamba effectively establishes long-distance dependencies. The Mamba model, built on selective SSM, has demonstrated considerable potential in natural language processing tasks. In recent studies, Chen et al. [35] applied Mamba to multispectral change detection using multiple stacked VMambas [36], which approach has proven effective for binary change detection, semantic change detection, and building damage assessment, showcasing the strong performance of Mamba in change detection tasks. Zhao et al. [37] further enhanced the selective scanning SSM by incorporating multiple scanning directions, enabling Mamba to capture more comprehensive spatial information and leading to superior results in multispectral semantic segmentation and change detection tasks. Dong et al. [38] introduced a method for parallel interaction between CNNs and Mamba, where the effective local modeling capabilities of CNNs and the efficient global modeling capabilities of Mamba complement each other to extract richer and more comprehensive features. Unlike multispectral images, HSIs contain a wealth of spectral information. To exploit this, Li et al. [39] proposed an improvement to Mamba in the spatial domain by dividing the HSI into multiple tokens along the spectral dimension, using Mamba in groups to explore correlations within the spectral dimension. However, this spectral grouping approach only models the spatial dimension of each token, limiting its ability to fully capture the information and correlations across the spectral dimension. To address this limitation, the improved method uses Mamba to simultaneously model the spatial and spectral dimensions of HSIs, effectively capturing their interdependencies.

In a previous research on change detection, most methods have primarily focused on overall detection accuracy. However, in image classification tasks, it is crucial to consider not only the overall accuracy (OA) but also the per-class accuracy and the average accuracy (AA) across all the categories. Analogous to classification tasks, multiclass change detection not only distinguishes between changed and unchanged regions but also identifies the category of each changed region. In practical remote sensing scenarios, small-sample categories typically correspond to critical yet infrequent land-cover changes, such as ecological disturbances, early-stage urban expansion, or subtle environmental alterations. Accurately detecting and classifying these small-sample categories is of paramount importance for effective resource management, ecological monitoring, and urban planning. Nevertheless, the classification performance for small-sample regions is typically suboptimal. This can be attributed to factors such as sample scarcity, small spatial scale, class imbalance, and strong noise interference.

Recent advances have addressed the challenges of small-sample classification through several strategies. Data augmentation techniques, for example, enhance the representation of small-sample classes to balance the distribution of categories and improve the detection performance for small-sample regions [40], [41]. Furthermore, well-designed loss functions, such as class-balanced or focal loss, can mitigate the issue of class imbalance and increase the model's focus on small targets [42], [43]. In addition, advanced deep learning architectures have been developed to improve the feature

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representation and learning of small target regions [44], [45]. Among these, wavelet transform has been widely used in classification and small object detection tasks due to its ability to capture multiscale and fine-grained features [46], [47], [48], [49].

To enhance the change detection performance and improve edge detection, a wavelet transform is introduced to convert the feature domain of the image into the frequency domain. In the wavelet domain [50], [51], most information is concentrated in the low-frequency components, while the high-frequency components primarily contain texture details and noise. Due to the one-way modeling characteristics of the standard SSM, it is less sensitive to noise. Drawing inspiration from this, the wave-Mamba attention module is proposed, which uses both SSM and the attention mechanism to extract features in different frequency domains, improving the robustness and effectiveness of the model in change detection. Currently, large language models are widely used in remote sensing vision [52]. To enhance the ability of the model to detect changes along image edges, a vision-based model, the segment anything model (SAM) [53], is incorporated. Recently, SAM has gained widespread use in various computer vision tasks. Trained on a large dataset of natural images, SAM exhibits strong generalization capabilities; it effectively handles objects and images it has not encountered during training. SAM has also been applied to change detection tasks. Wang et al. [54] proposed to combine class activation mapping (CAM) with SAM to generate pseudolabels for change detection. SAM was initially used in weakly supervised change detection, where it showed promising results. Ding et al. [55] leveraged the robust semantic recognition abilities of SAM to improve its capability to learn semantic information while simultaneously reducing reliance on large amounts of training data. SAM is typically used to generate pseudolabels and enhance semantic information in change detection. In the current approach, SAM is extended to HSI. Specifically, the powerful segmentation capabilities of SAM are used to generate a binary mask for HSI, which is then used to improve the ability of the model to detect changes, particularly along the edges of the HSI.

Through the above design, this method not only improves the overall detection but also improves the detection of each category. The main contributions of this work are as follows.

1) An encoder based on the spectral-guided spatial-enhanced Mamba (SSEM) block is proposed to capture spatial–spectral features across multiple scales. This design effectively encodes HSIs, ensuring better adaptability for downstream decoding tasks.
2) A knowledge-guided difference block inspired by the SAM is introduced and extended to HSI domain. By leveraging the binary mask generated by the SAM, the representation of change features is enhanced, significantly improving model accuracy.
3) To enhance feature fusion, a multiscale fusion decoder incorporating the wave-Mamba attention module is developed. By integrating the wavelet transform with the SSM, this module not only improves small-sample category detection but also strengthens edge detection for more precise change identification.

Experiments are conducted on multiple widely recognized HSI multiclass change detection datasets. Compared with other advanced HSI-CD methods, the proposed method shows very competitive, even SOTA, performance.

The rest of this article is organized as follows. In Section II, we introduce the proposed spatial–spectral feature-enhanced Mamba and SAM-guided (SFMS). Section III describes the experimental settings and the results of the experiment on the public datasets. The ablation experiments for the model and parameters are given in Section IV. Finally, Section V summarizes the conclusions.

II. Methodology

A. Preliminaries

Benefiting from continuous-time systems, SSM methods have recently emerged. In the SSM model, a 1-D function or sequence $x(t) \in \mathbb{R}$ is mapped to $y(t) \in \mathbb{R}$ through a hidden state $h(t) \in \mathbb{R}^N$, which is governed by the following linear ordinary differential equation (ODE):

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{aligned} \tag{1}$$

where $A \in \mathbb{R}^{N\times N}$ is the state matrix that stores all the previous historical information, and $B \in \mathbb{R}^{N\times 1}$ and $C \in \mathbb{R}^{N\times 1}$ represent projection parameters.

Most modern deep learning frameworks and training algorithms, such as gradient descent and backpropagation, are designed for discrete-time models. To directly integrate SSM into current deep learning algorithms, the zero-order hold (ZOH) method is used to discretize the continuous parameters A and B in SSM. An appropriate time step parameter Δ is introduced in the discretization process

$$\begin{aligned} \bar{A} &= \exp(\Delta A) \\ \bar{B} &= (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B. \end{aligned} \tag{2}$$

After discretization, the final output can be directly obtained by a full convolution calculation. According to (1) and (2), it can be written as

$$\begin{aligned} h(t) &= \bar{A}h_{t-1} + \bar{B}x_t \\ y(t) &= Ch_t. \end{aligned} \tag{3}$$

Because the parameters in the above SSM do not change with different inputs, the recent Mamba combines the scanning mechanism with the learnable parameters B, C, and Δ, making B, C, and Δ functions of the input. This allows the model to adaptively adjust its behavior based on the input content, enabling the SSM to dynamically learn contextual information. In addition, a hardware-aware optimization is proposed to ensure computational efficiency.

B. Overview

An overview of the proposed method is shown in Fig. 1. The framework consists of three main components: a multiscale spatial–spectral Mamba encoder, an SAM-based prior

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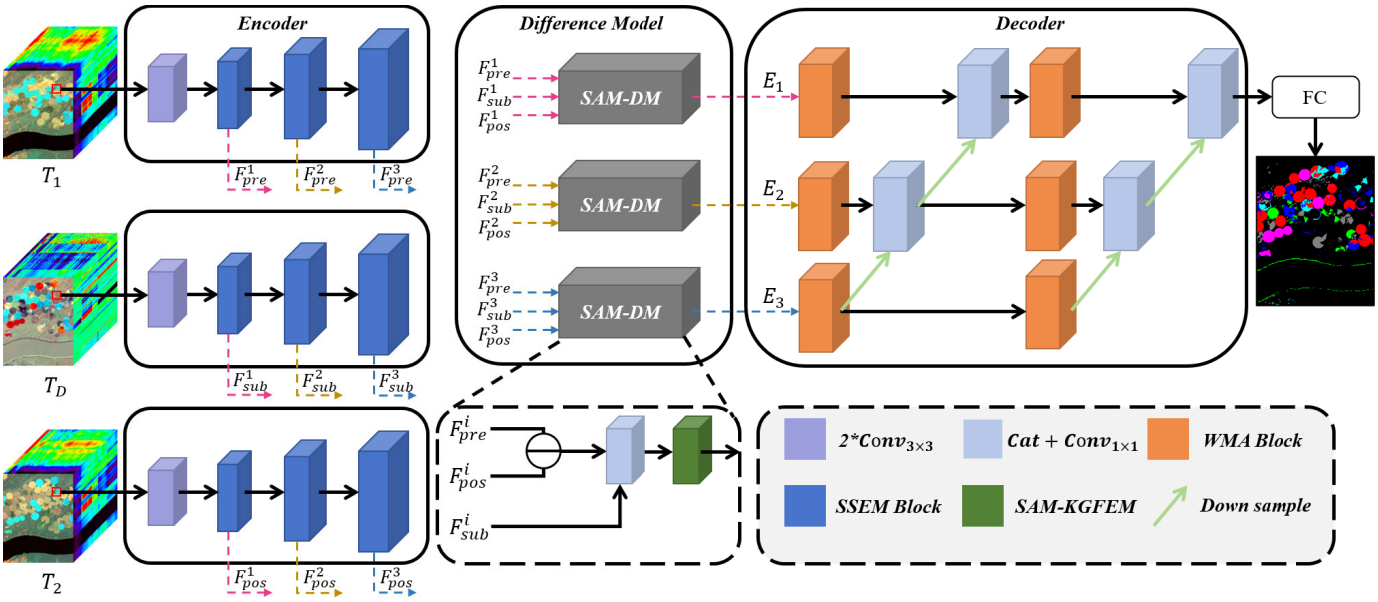


Fig. 1. SFMS structure. The gray area in the lower right corner is the diagram annotation.

knowledge-guided difference module, and a multiscale feature fusion wavelet Mamba decoder.

Given a dual-temporal HSI patch pair, $T_1 \in \mathbb{R}^{C\times H\times W}$ and $T_2 \in \mathbb{R}^{C\times H\times W}$, along with their difference image $T_D \in \mathbb{R}^{C\times H\times W}$, where C represents the channel dimension, and H and W represent the height and width of the images, respectively. The input patches are first processed by the multiscale spatial–spectral Mamba encoder. This encoder, using shared weights, extracts the feature maps $F^i{}_{\text{pre}}$, $F^i{}_{\text{pos}}$, and $F^i{}_{\text{dif}} \in \mathbb{R}^{2^{i-1}\times h\times 2^{i-1}\times w\times(c/2^{i-1})}$, for $i = 1, 2, 3$, where c is the embedding dimension, and h and w correspond to the height and width of the feature maps at different scales, respectively.

The extracted feature maps are then passed into the SAM prior knowledge-guided difference module, which performs a feature fusion operation to highlight the areas of change, thereby enhancing the representation of these regions. This results in updated feature maps $E_i \in \mathbb{R}^{2^{i-1}\times h\times 2^{i-1}\times w\times(c/2^{i-1})}$, for $i = 1, 2, 3$.

Finally, the fused features E_i are input into the multiscale feature fusion wavelet Mamba decoder block. This block aggregates information from multiple scales within the same layer to better capture contextual features. The output is then processed through a multilayer perceptron (MLP), generating the final multiclass change detection map.

C. Multiscale Spatial–Spectral Mamba Encoder

As shown in Fig. 1, a three-branch Siamese architecture is used, which differs from the conventional design. The dual-phase images, along with their difference image, are fed into the multiscale spatial–spectral Mamba encoder. The difference image is particularly useful for learning the changed areas in the image pair, allowing the model to focus on the changed regions. Initially, the patch pairs T_1 and T_2 from the dual-phase images along with their corresponding difference image T_D are passed through a convolution block with multiple convolution kernels to extract shallow features. This results in $f_{\text{pre/pos/sub}} \in \mathbb{R}^{h\times w\times c}$, where h, w, and c represent the height, width, and channel dimensions, respectively. These extracted features are then input into the SSEM to capture features at multiple scales. The formulas for this process are as follows:

$$\begin{aligned} f^1{}_{\text{pre/pos/sub}} &= \text{Stem}(T) \\ F^1{}_{\text{pre/pos/sub}} &= \text{SSEM}\left(f^1{}_{\text{pre/pos/sub}}\right) \\ F^2{}_{\text{pre/pos/sub}} &= \text{SSEM}\left(\text{Up}\left(F^1{}_{\text{pre/pos/sub}}\right)\right) \\ F^3{}_{\text{pre/pos/sub}} &= \text{SSEM}\left(\text{Up}\left(F^2{}_{\text{pre/pos/sub}}\right)\right) \end{aligned} \tag{4}$$

where Stem$(\cdot)$ consists of two 3×3 convolution operations, SSEM$(\cdot)$ is the spectrally guided spatial-enhanced Mamba block, and UP$(\cdot)$ is the upsample operation. In HSI change detection, the spectral information of the image plays a crucial role in ensuring accurate detection. The objective of this work is to explore an effective Mamba structure that integrates both the spatial and spectral information. Traditionally, Mamba methods focus on scanning the spatial dimensions of the image. A straightforward extension of this approach is to scan the spectral dimension to capture its spectral features. Taking the first block of the T_1 branch as an example, as illustrated in Fig. 2(a), the SSEM consists of three branches. The inputs for the other two branches are the transposed versions of the feature maps $f^{T_a}_{\text{pre}} \in \mathbb{R}^{h\times c\times w}$ and $f^{T_b}_{\text{pre}} \in \mathbb{R}^{w\times c\times h}$. Initially, layer normalization is applied to all the three branches before they are modeled with the SSM to capture the long-range dependencies within the spatial and spectral dimensions. Fig. 2(b) illustrates the scanning and modeling process. f_{pre}, $f^{T_a}_{\text{pre}}$, and $f^{T_b}_{\text{pre}}$ are flattened to form $\text{Spa} \in \mathbb{R}^{\text{hw}\times c}$, $\text{Spe}_1 \in \mathbb{R}^{\text{hc}\times w}$, and $\text{Spe}_2 \in \mathbb{R}^{\text{wc}\times h}$. Subsequently, Spa, Spe_1, and Spe_2 are each modeled using the proposed SSM, as defined in (3). S_1, S_1', and S_1'' are then reshaped into tensors for subsequent processing. Next, the outputs S_1' and S_1'' are weighted and fused. The learnable parameters ω_1 and ω_2 are updated during training. The fused result is then passed through a sigmoid function

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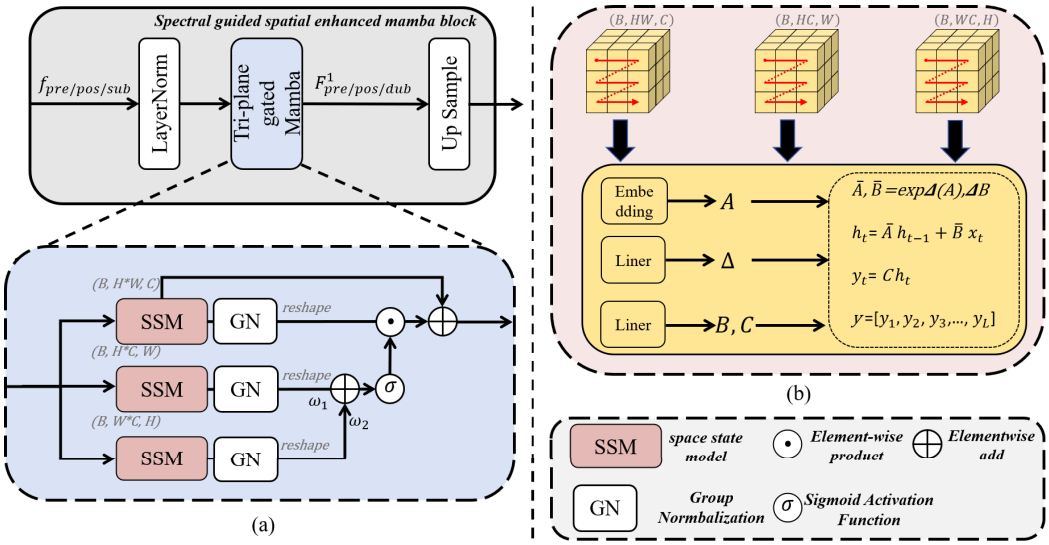


Fig. 2. SSEM block structure. (a) Structure diagram of the tri-Mamba. (b) Flowchart for scanning and modeling the three planes in (a). The gray part in the lower right corner is the annotation.

and multiplied by S_1. This allows the spectral information extracted by the SSM to enhance the spatial information of S_1. Finally, the residual of S_1 is added to the result. This process is expressed as follows:

$$\begin{aligned}
\mathrm{Spa} &= \mathrm{Flatten}\left(\mathrm{LN}\left(f_{\mathrm{pre}}\right)\right) \\
S_1 &= \mathrm{GN}(\mathrm{SSM}(\mathrm{Spa})) \\
\mathrm{Spe}_1 &= \mathrm{Flatten}\left(\mathrm{LN}\left(f_{\mathrm{pre}}^{T_a}\right)\right) \\
S_1' &= \mathrm{GN}\left(\mathrm{SSM}\left(\mathrm{Spe}_1\right)\right) \\
\mathrm{Spe}_2 &= \mathrm{Flatten}\left(\mathrm{LN}\left(f_{\mathrm{pre}}^{T_b}\right)\right) \\
S_1'' &= \mathrm{GN}\left(\mathrm{SSM}\left(\mathrm{Spe}_2\right)\right) \\
f_{\mathrm{pre}}^1 &= S_1 \cdot \mathrm{Sig}\left(\omega_1 S_1' + \omega_2 S_1''\right) + S_1
\end{aligned} \tag{5}$$

where $\mathrm{LN}(\cdot)$ represents the layer normalization operation, $\mathrm{SSM}(\cdot)$ represents the SSM function, and $\mathrm{Sig}(\cdot)$ stands for the sigmoid activation function.

Next, f_{pre}^1 is upsampled through the same SSEM block to obtain feature maps at different scales. Unlike traditional multiscale methods that rely on downsampling, an upsampling approach is used here. Downsampling often requires adjusting the number of channels in the feature map to prevent information loss, which increases model complexity. To avoid this, upsampling is used to extract multiscale information while maintaining model simplicity, ensuring that the accuracy is not compromised.

D. SAM Prior Knowledge-Guided Difference Module

The feature maps at each scale are input into the difference module. In this module, the corresponding dual-phase feature image pair is subtracted, and the result is concatenated with the difference image. To reduce the number of channels, a 1×1 convolution is applied, resulting in feature maps at three different scales. The process can be expressed as follows:

$$E_1 = \mathrm{Conv}\left(\mathrm{Cat}\left(\left(f_{\mathrm{pre}}^1 - F_{\mathrm{pos}}^1\right), f_{\mathrm{sub}}^1\right)\right)$$

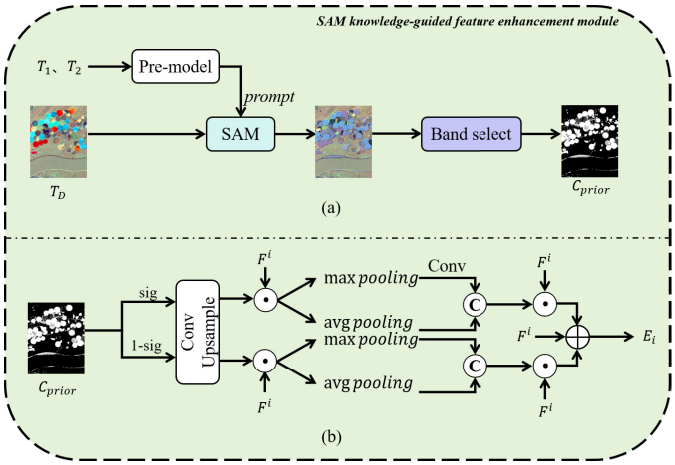


Fig. 3. SAM-KGFEM structure. (a) Structure of the SAM segmentation change detection prior mask. (b) Structure for enhancing the features of the changed regions using the prior mask.

$$\begin{aligned}
E_2 &= \mathrm{Conv}\left(\mathrm{Cat}\left(\left(f_{\mathrm{pre}}^2 - f_{\mathrm{pos}}^2\right), f_{\mathrm{sub}}^2\right)\right) \\
E_3 &= \mathrm{Conv}\left(\mathrm{Cat}\left(\left(f_{\mathrm{pre}}^3 - f_{\mathrm{pos}}^3\right), f_{\mathrm{pos}}^3\right)\right)
\end{aligned} \tag{6}$$

where $\mathrm{Cat}(\cdot)$ is the concat operation, which stacks the feature maps together in the channel dimension, and $\mathrm{Conv}(\cdot)$ is the convolution operation. As shown in Fig. 3, a mask is introduced as prior knowledge to enhance the representation of change features. Initially, unsupervised two-class change detection methods are considered to generate masks for the HSI. However, these methods tend to perform poorly when detecting categories with few samples in multiclass settings. Therefore, the binary mask generated by the unsupervised method [56] is used as a prompt for the SAM. Due to the specific nature of the SAM pretrained model, which only accepts three-channel images for segmentation, pseudo-three-channel images are generated by repeating the channels of the original HSI.

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While SAM may not perform optimally for remote sensing image segmentation, the generated mask can be improved by selecting the most relevant bands through a band selection process. In HSIs, each band contains different information. Therefore, for each band the segmentation mask corresponding to the pseudo three channels also differs significantly. To obtain the best mask map, the mask map of each pseudo-three-channel is filtered through a band selection process. Specifically, for each pixel at position (x, y), if its label and mask values are greater than 0, the pixel is considered to contain useful information and is counted. This counting occurs across all the channels. The image is sorted based on the count values, and the band that best represents the change feature is selected using a threshold T. Finally, the mask images corresponding to these selected bands are merged. The counting formula is as follows:

$$\text{Count}_C = \begin{cases} \text{Count}_C + 1, & \text{where mask}_C^{x,y} > 0 \\ & \text{and label}_{x,y} > 0 \\ \text{Count}_C & \text{else} \end{cases} \tag{7}$$

where C is the number of channels, mask_C is the mask of the Cth channel, Count_C is the count value of the Cth channel mask, x, y are the pixel coordinates, and label is the training sample.

Then, its scale is adapted to the corresponding feature map, and knowledge-guided feature enhancement module (KGFEM) [57] is used to enhance E_i.

E. Multiscale Feature Fusion Wavelet Mamba Decoder

To effectively encode and decode contextual information, a common approach in CNNs is to gradually reduce the size of the feature map, extracting deeper and more semantically meaningful low-resolution features. Then, the feature map is progressively restored from the low-resolution representation to a high-resolution one. Such multiscale mechanisms for collecting spatial information are often more effective in CNNs. Building on this concept, this work proposes a multiscale feature fusion wavelet Mamba decoder, as illustrated in Fig. 1. After obtaining E_i, it is input into the decoder, and then the decoder fuses the different scales at the same level, which are subsequently passed on to the next level as follows:

$$\begin{aligned} E_i^{\text{wma}} &= \text{WMA}(E_i) \\ E_3^k &= E_3^{\text{wma}} \\ E_2^k &= \text{Conv}\left(\text{Cat}\left(E_2^{\text{wma}}, \text{Down}\left(E_3^{\text{wma}}\right)\right)\right) \\ E_1^k &= \text{Conv}\left(\text{Cat}\left(E_1^{\text{wma}}, \text{Down}\left(E_2^1\right)\right)\right) \end{aligned} \tag{8}$$

where $\text{WMA}(\cdot)$ is the wavelet Mamba attention block (WMAB) operation, k is the level, $\text{Down}(\cdot)$ is the downsampling operation, $\text{Cat}(\cdot)$ is the concatenation operation, and $\text{Conv}(\cdot)$ is a convolution operation with a convolution of 1.

The WMAB is shown in Fig. 4. First, the discrete wavelet transform (DWT) is used to decompose E_1 into high-frequency information E_1^{HL}, E_1^{LH}, and E_1^{HH} and low-frequency information, E_1^{LL}, which is expressed as

$$E_1^{\text{HL}}, E_1^{\text{LH}}, E_1^{\text{HH}}, E_1^{\text{LL}} = \text{DWT}(E_1). \tag{9}$$

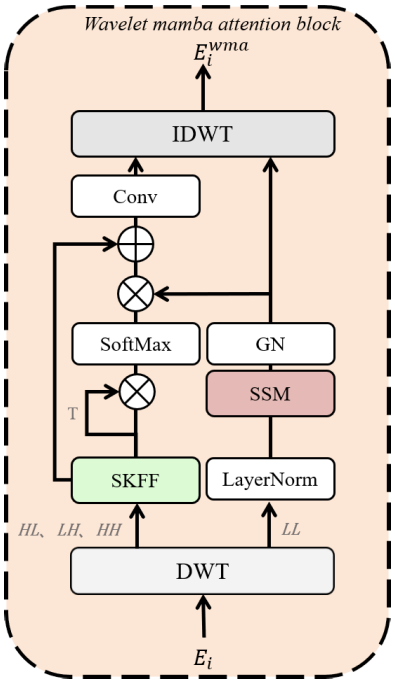


Fig. 4. Structure of the WMAB.

Because the low-frequency domain contains relatively rich image information, the low-frequency information E_1^{LL} is modeled using the SSM. In contrast, the high-frequency domain contains limited texture information and significant noise [60], making it unsuitable for modeling with SSM. To address this, E_1^{HL}, E_1^{LH}, and E_1^{HH} are fused using the SKFF method [61]. To further enhance the model's ability to detect texture features, an attention mechanism is applied to strengthen the fused high-frequency information by incorporating the low-frequency information modeled by SSM. Finally, the feature map is reconstructed using an inverse DWT (IDWT). The process can be expressed as follows:

$$\begin{aligned} E_1^{\text{LS}} &= \text{SSM}\left(\text{LN}\left(E_1^{\text{LL}}\right)\right) \\ E_1^H &= \text{SKFF}\left(E_1^{\text{HH}}, E_1^{\text{HL}}, E_1^{\text{LH}}\right) \\ E_1^L &= \text{soft max}\left(\frac{E_1^{\text{LS}} \times \left(E_1^{\text{LS}}\right)^{\text{Trans}}}{\sqrt{C}}\right) \\ E_1^H &= E_1^H + E_1^L \times E_1^H \\ E_1 &= \text{IDWT}\left(\text{Conv}\left(E_1^H\right), E_1^{\text{LS}}\right) \end{aligned} \tag{10}$$

where $\text{LN}(\cdot)$ denotes the layer normalization operation.

After two levels of multiscale feature fusion, the final feature map is passed through the classification head with two fully connected layers to generate the final multiclass change detection map.

III. Experimental Results and Analysis

A. Datasets

The datasets used in these experiments were collected by the Earth observation-1 (EO-1) satellite, which is equipped with the hyperion hyperspectral sensor. Fig. 5 illustrates the China, USA, and Hermiston datasets. The spatial resolution of

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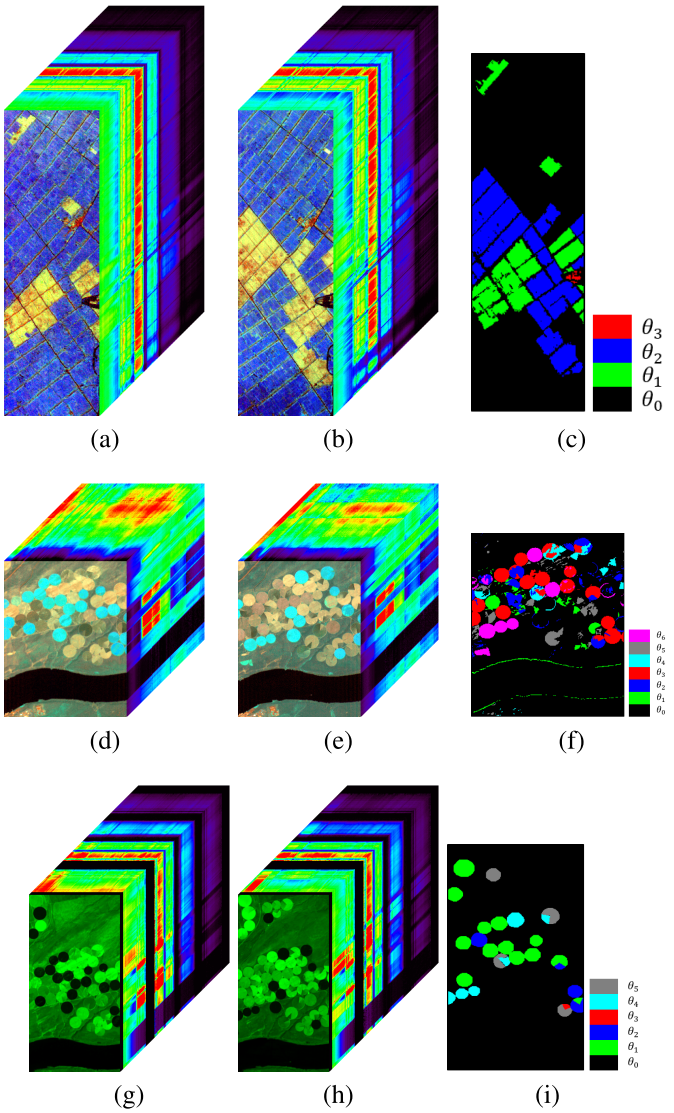


Fig. 5. HSI change detection datasets. (a)–(c) China dataset. (d)–(f) USA dataset. (g)–(i) Hermiston dataset.

the HSIs captured by the EO-1 sensor is approximately 30 m, while the spectral resolution is 10 nm.

The China dataset covers farmland near Yancheng, Jiangsu Province, China. The prephase image Fig. 5(a) was taken on May 3, 2006, and the postphase image Fig. 5(b) was captured on April 23, 2007. Fig. 5(c) shows the multiclass reference change map. The size of this dataset is 450 × 140 and has 155 bands. Three change classes are included in this dataset, as shown in Table I.

The USA dataset covers irrigated farmland in Umatilla County, Oregon, USA. The prephase image Fig. 5(d) was captured on May 1, 2004, and the postphase image Fig. 5(e) on May 8, 2007. Fig. 5(f) shows the multiclass reference change map. The dataset image contains a size of 307 × 241 and 154 bands. Its land-cover types are irrigated fields, soils, grasslands, rivers, cropland, and buildings. Table I shows specific information for each of its change categories.

The Hermiston dataset includes farmland in Hermiston, Oregon, USA, with the prephase image Fig. 5(g) taken in 2004 and the postphase image Fig. 5(h) in 2007. Fig. 5(i) shows the multiclass reference change map. The spatial dimension of the dataset image is 390 × 200 and contains 242 bands. This dataset contains five types of changes related to crop transitions, as detailed in Table I.

TABLE I
INFORMATION ON THE DATASETS

Class	Number of change pixels		
	China	USA	Hermiston
0	44723	57260	68014
1	6559	1850	5558
2	11613	2905	1331
3	105	5896	79
4		1524	1557
5		2744	1461
6		1808	
Total	63000	73987	78000
Train	12600	14797	15600

B. Experimental Settings

The PyTorch framework was used to implement the proposed method, and models were run on a single NVIDIA RTX 2070 TI GPU. The Adam optimizer [62] with L2 regularization was used, with both the learning rate and weight decay set to 0.0001. The batch size was configured to 32, and the patch size was set to 7. In these experiments, 20% of the samples from each category were randomly selected as training data. The sample size of each dataset used for training is shown in Table I. The experiments were repeated three times, and the average results of the OA, AA, and Kappa coefficient were reported. To validate the effectiveness of our method, it was compared with seven classic and state-of-the-art approaches, including CVA+MLP [16], HybridSN [28], MMSRC [29], CSANet [30], SFFGL [58], MFCEN [59], and MambaHSI [39]. The evaluation metrics used to assess the methods were OA, AA, and the Kappa coefficient.

C. Change Detection Results on the China Dataset

Fig. 6 and Table II present the results of multiclass change detection on the China dataset. While current methods achieve good overall performance in multiclass detection, the detection results for smaller area categories in the China dataset are suboptimal, with many missed and false detections. This is particularly evident for the category labeled "3," where the training samples account for only 20% of each category, leading to a small-sample size that hinders effective feature representation learning. Among the methods, CVA+MLP fails to recognize this small-sample category, while CSANet and MFCEN exhibit numerous false detections. However, due to the feature enhancement provided by SAM-KGFEM for detecting changed areas and the powerful feature modeling ability of WMAB for color and texture, the proposed method shows minimal false detections and achieves good detection results for this category. Its detection performance for this

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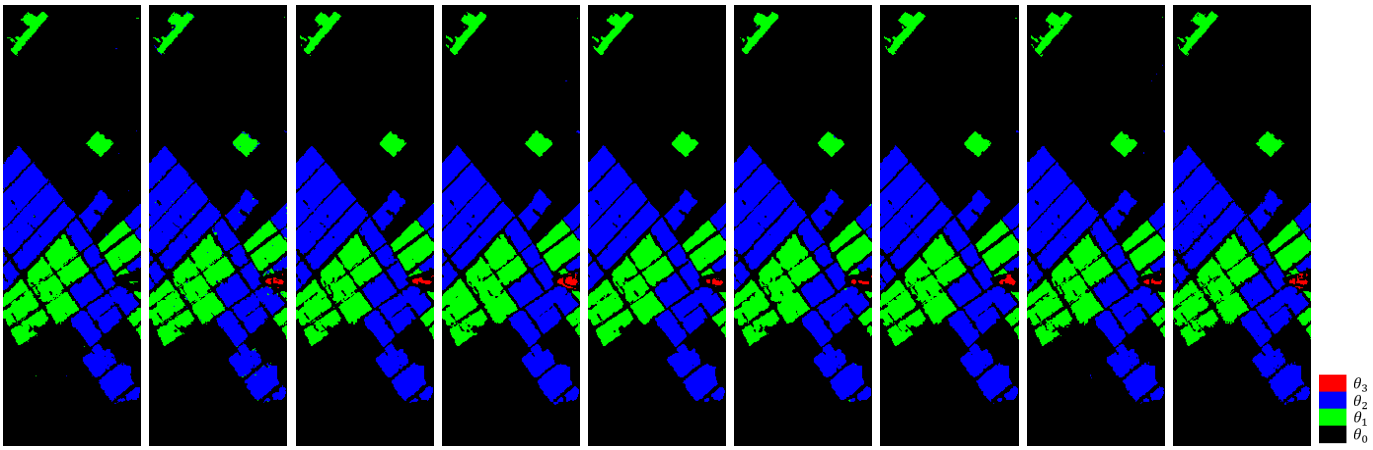


Fig. 6. Visualization of the proposed and comparative methods on the China dataset. (a) CVA + MLP. (b) HybridSN. (c) MMSRC. (d) CSANet. (e) SFFGL. (f) MambaHSI. (h) MFCEN. (h) This work. (i) Ground truth.

TABLE II
RESULTS OF COMPARATIVE EXPERIMENTS ON THE CHINA DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUE HIGHLIGHTED IN RED AND THE SECOND HIGHEST IN BLUE

class	CVA+MLP [16]	HybridSN [28]	MMSRC [29]	CSANet [30]	SFFGL [58]	MambaHSI [39]	MFCEN [59]	Ours
0	98.52±0.16	98.80±0.07	98.85±0.03	98.80±0.05	98.47±0.03	98.63±0.02	98.88±0.02	98.85±0.04
1	95.68±0.38	95.47±0.11	96.88±0.02	94.15±0.12	95.54±0.28	96.48±0.14	96.36±0.17	96.50±0.21
2	97.01±0.29	97.67±0.03	97.22±0.07	96.17±0.14	95.96±0.10	97.33±0.05	98.01±0.03	97.08±0.13
3	0.00±0.00	84.33±1.12	83.78±3.34	62.06±5.82	73.03±6.12	88.89±2.19	70.52±4.52	96.00±0.57
OA	97.94±0.21	98.22±0.15	98.28±0.09	97.72±0.11	97.66±0.08	98.16±0.06	98.22±0.04	98.27±0.03
AA	72.80±0.20	94.07±0.33	94.18±0.86	87.79±1.53	90.75±1.63	95.33±0.60	90.95±1.18	97.11±0.23
Kappa	95.44±0.36	96.06±0.12	96.02±0.08	95.00±0.19	94.84±0.13	95.92±0.10	95.50±0.11	96.18±0.06

category is 7.11 higher than that of the second-best method. There is no significant improvement as the OA and Kappa values are not a good representation of the small categories of enhancement. Although the OA of the proposed method is not the highest, its AA and Kappa coefficient surpass the second-best method by 1.78 and 0.12, respectively, demonstrating the effectiveness of this approach.

D. Change Detection Results on the USA Dataset

The multiclass detection results for the proposed method and other methods on the USA dataset are shown in Fig. 7. This dataset contains numerous small-change regions, and the categories are adjacent. The inherent complexity of the dataset leads to suboptimal multiclass change detection results across various methods. Methods such as CVA+MLP, HybridSN, CSANet, SFFGL, and MFCEN struggle with missed detections, particularly along the river edge. However, due to the powerful global modeling and featured extraction capabilities of the tri-plane gated Mamba (tri-Mamba) and WMAB, the proposed method detects these small changes relatively well. As shown in Table III, although this method does not achieve the best detection results for every category, its AA is still 0.2 points higher than the second-best method, demonstrating the effectiveness of our approach.

E. Change Detection Results on the Hermiston Dataset

The multiclass detection results of the proposed method, along with those of other comparison methods, are shown for the Hermiston dataset in Fig. 8. This dataset is relatively simple with clear textures but it presents the challenge of small-sample categories. Among the methods, CVA+MLP struggles to detect the small-sample category. As shown in Table IV, the proposed method achieves the highest detection performance for the small-sample category with label "3," surpassing the second-best method by 3.05 points. This highlights the effectiveness of SAM-KGFEM in enhancing the features of the changed areas and the powerful modeling capabilities of WMAB for features such as color and texture. In addition, our method achieves the best AA, with 1.12 points higher than the second-best result, further demonstrating its robustness.

IV. DISCUSSION

A. Parameter Quantity Analysis

To evaluate the effectiveness of the proposed model, we conducted a comprehensive comparison in terms of parameter count (Params), floating-point operations (FLOPs), OA, and AA, as summarized in Table V. From the results, although our method incurs a computational cost of 5.46 GFLOPs,

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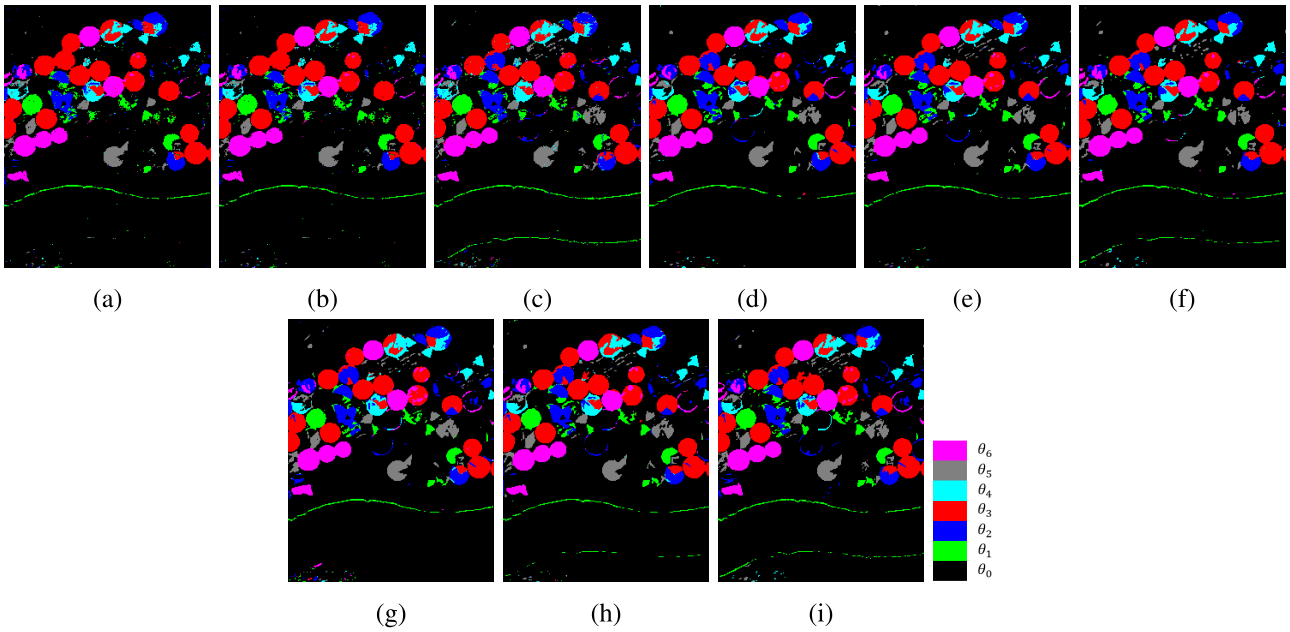


Fig. 7. Visualization of the proposed and comparative methods on the USA dataset. (a) CVA + MLP. (b) HybridSN. (c) MMSRC. (d) CSANet. (e) SFFGL. (f) MambaHSI. (h) MFCEN. (h) This work. (i) Ground truth.

TABLE III

RESULTS OF COMPARATIVE EXPERIMENTS ON THE USA DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUE HIGHLIGHTED IN RED AND THE SECOND HIGHEST IN BLUE

class	CVA+MLP [16]	HybridSN [28]	MMSRC [29]	CSANet [30]	SFFGL [58]	MambaHSI [39]	MFCEN [59]	Ours
0	96.42±1.36	98.26±0.16	98.84±0.06	97.54±0.23	97.94±0.11	97.95±0.09	98.28±0.03	97.97±0.12
1	77.26±3.01	81.34±1.39	87.96±0.88	86.20±1.31	85.16±1.80	85.71±0.86	89.79±0.41	90.13±0.32
2	86.96±1.47	81.93±1.43	82.43±0.72	88.08±0.16	89.86±0.05	88.46±0.23	87.91±0.19	88.94±0.17
3	88.50±1.03	93.91±0.87	95.93±0.09	96.23±0.19	96.63±0.07	96.02±0.13	98.68±0.05	95.65±0.14
4	84.95±2.06	80.09±2.05	83.87±0.57	86.67±0.57	89.09±0.03	84.24±0.77	83.17±0.37	89.03±0.05
5	95.58±0.51	96.28±0.12	95.54±0.08	95.63±0.09	96.40±0.10	94.50±0.08	91.94±0.06	95.76±0.08
6	86.40±1.09	79.76±1.61	87.62±0.05	88.67±0.06	88.15±0.15	90.40±0.07	86.07±0.09	87.14±0.10
OA	94.63±1.19	95.98±0.90	96.99±0.11	96.36±0.18	96.58±0.07	96.58±0.13	96.74±0.05	96.77±0.08
AA	88.01±1.50	87.37±1.09	90.31±0.35	91.29±0.37	91.89±0.33	91.04±0.31	90.83±0.17	92.09±0.14
Kappa	85.69±2.16	89.63±0.83	92.27±0.17	90.49±0.29	91.62±0.11	91.14±0.19	91.52±0.10	91.59±0.16

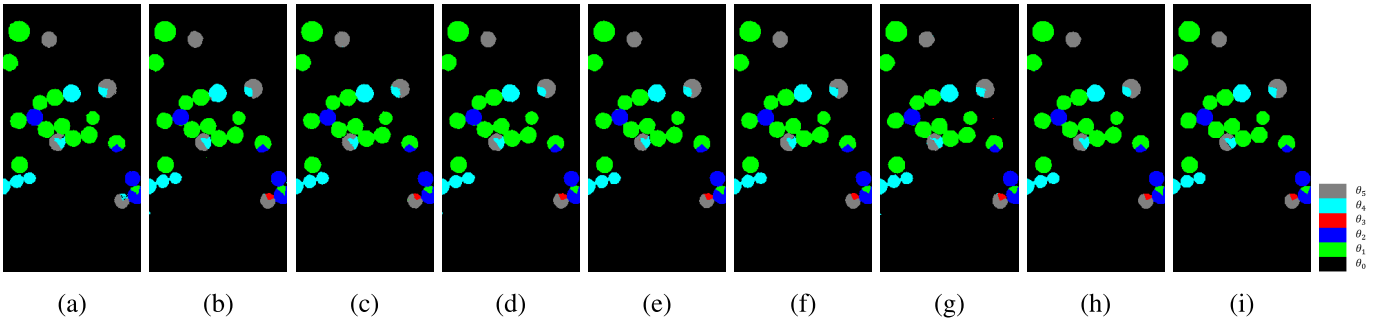


Fig. 8. Visualization of proposed method and comparative methods on the Hermiston dataset. (a) CVA + MLP. (b) HybridSN. (c) MMSRC. (d) CSANet. (e) SFFGL. (f) MambaHSI. (h) MFCEN. (h) This work. (i) Ground truth.

it achieves the highest AA score among all the compared methods. While our OA score is not the highest, it is noteworthy that our model requires fewer FLOPs than the MMSRC method, which achieved the top OA score. This demonstrates the efficiency and competitive performance of our approach in balancing accuracy and computational complexity.

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TABLE IV

RESULTS OF COMPARATIVE EXPERIMENTS ON THE HERMISTON DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUE HIGHLIGHTED IN RED AND THE SECOND HIGHEST IN BLUE

class	CVA+MLP [16]	HybridSN [28]	MMSRC [29]	CSANet [30]	SFFGL [58]	MambaHSI [39]	MFCEN [59]	Ours
0	99.76±0.09	99.72±0.08	99.64±0.05	99.69±0.06	99.66±0.04	99.70±0.03	99.77±0.04	99.70±0.03
1	97.11±0.29	97.59±0.13	97.79±0.07	97.67±0.10	97.97±0.07	98.38±0.06	97.80±0.09	97.83±0.07
2	97.88±0.13	97.11±0.10	96.81±0.08	97.06±0.11	97.49±0.06	96.71±0.08	98.01±0.06	97.49±0.05
3	0.00±0.00	88.63±1.44	94.28±0.91	89.65±1.13	84.78±1.77	89.15±1.49	79.79±2.92	97.33±0.09
4	91.98±0.82	95.72±0.41	93.78±0.35	95.26±0.39	96.44±0.22	95.97±0.15	96.17±0.14	96.25±0.12
5	96.23±0.24	95.06±0.26	96.06±0.12	96.07±0.17	97.60±0.10	95.78±0.13	95.53±0.11	96.47±0.09
OA	99.31±0.09	99.34±0.05	99.27±0.04	99.33±0.06	99.39±0.02	99.39±0.03	99.42±0.07	99.40±0.05
AA	80.49±0.26	95.64±0.23	96.39±0.26	95.90±0.32	95.66±0.37	95.95±0.32	94.51±0.56	97.51±0.07
KAPPA	97.05±0.19	97.19±0.11	96.89±0.09	97.15±0.12	97.39±0.08	97.43±0.05	97.55±0.10	97.41±0.09

TABLE V

COMPARISON OF COMPUTATION COST ON THE CHINA DATASET

Metrics	CVA+MLP [16]	HybridSN [28]	MMSRC [29]	CSANet [30]	SFFGL [58]	MambaHSI [39]	MFCEN [59]	Ours
FLOPs (G)	0.15	0.38	14.49	2.85	1.21	0.04	9.90	5.46
Params (M)	0.11	0.62	0.07	2.43	6.83	0.07	76.78	2.22
OA (%)	97.94±0.21	98.22±0.15	98.28±0.09	97.72±0.11	97.66±0.08	98.16±0.06	98.22±0.04	98.27±0.03
AA (%)	72.80±0.20	94.07±0.33	94.18±0.86	87.79±1.53	90.75±1.63	95.33±0.60	90.95±1.18	97.11±0.23

TABLE VI

ABLATION EXPERIMENT RESULTS FOR THE CHINA, USA, AND HERMISTON DATASETS. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUE HIGHLIGHTED IN RED

Tri-mamba	SAM	WMAB	China			USA			Hermiston		
			OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
			98.11±0.12	71.22±0.17	95.56±0.20	96.26±0.15	91.01±0.33	90.39±0.29	99.27±0.14	96.52±0.21	96.89±0.33
✓			98.16±0.08	93.04±0.27	95.86±0.19	96.46±0.17	91.22±0.25	90.97±0.28	99.30±0.11	96.68±0.20	97.02±0.29
	✓		98.21±0.09	96.46±0.22	96.03±0.25	96.57±0.16	91.01±0.34	91.05±0.27	99.32±0.10	96.72±0.17	97.09±0.16
		✓	98.19±0.14	96.86±0.29	95.98±0.18	96.52±0.15	91.20±0.22	90.97±0.30	99.31±0.08	97.15±0.14	97.06±0.15
✓	✓		98.26±0.06	96.58±0.19	96.10±0.16	96.70±0.14	91.39±0.19	91.43±0.20	99.33±0.10	96.88±0.24	97.13±0.26
✓		✓	98.23±0.07	93.18±0.30	96.07±0.13	96.69±0.13	91.32±0.18	91.42±0.23	99.34±0.08	96.27±0.26	97.16±0.14
	✓	✓	98.27±0.05	95.04±0.20	96.18±0.10	96.65±0.12	91.24±0.26	91.33±0.26	99.33±0.06	96.72±0.12	97.14±0.13
✓	✓	✓	98.27±0.03	97.11±0.23	96.18±0.06	96.77±0.08	92.09±0.14	91.59±0.16	99.40±0.05	97.51±0.07	97.41±0.09

B. Ablation Study

In the ablation study, it was replaced with the simplest plane-scanning Mamba, removing the enhancement of the two planes in the spectral dimension. SAM refers to the difference module using SAM-KGFEM. In this study, for WMAB, we removed WMAB and replaced it with a self-attention mechanism.

From the analysis presented in Table VI, incorporating the two blocks, SAM-KGFEM and WMAB, improves the AA index across all the three datasets. This indicates that these two blocks significantly enhance the model's ability to capture fine-grained color and texture features. Furthermore, when the tri-Mamba is removed, there is a noticeable decline in the OA, AA, and Kappa values across all the three datasets. This underscores the critical role that the tri-Mamba plays in enhancing the model's capacity to capture global contextual information.

C. Effect of Tri-Mamba on Change Detection Results

An ablation study focusing on the multiplane scanning capabilities was performed to investigate the impact of the tri-Mamba on change detection results. As shown in Table VII, the influence of a tri-Mamba on the change detection outcomes for the China dataset is examined, where a, b, and c represent the state space models derived from the three plane scans of H-W, H-C, and W-C, respectively. While the OA achieves satisfactory results when modeled from the H-W plane alone, the AA performance is suboptimal due to insufficient spectral feature extraction. To address this, dual-plane modeling was used to enhance the extraction of spectral features. Specifically,

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TABLE VII

TRI-MAMBA ABLATION RESULTS FOR THE CHINA DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUE HIGHLIGHTED IN RED

a	b	c	China		
			OA	AA	Kappa
✓			98.27±0.08	95.04±0.29	96.18±0.12
✓	✓		98.27±0.09	96.40±0.33	96.17±0.17
✓		✓	98.25±0.06	96.04±0.25	96.13±0.14
✓	✓	✓	98.27±0.03	97.11±0.23	96.18±0.06

TABLE VIII

THRESHOLD T ABLATION RESULTS FROM THE CHINA DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUE HIGHLIGHTED IN RED

	T=0.5	T=0.3	T=0.1	T=0.05
OA	98.22±0.12	98.27±0.08	98.27±0.03	98.17±0.06
AA	93.05±0.32	95.05±0.28	97.11±0.23	73.24±0.19
Kappa	96.06±0.12	96.22±0.10	96.18±0.06	95.93±0.23

dual-plane models combining H-W and H-C, as well as H-W and W-C resulted in notable improvements in both OA and AA, with the most significant enhancement observed in AA. This improvement is attributed to each spectral plane capturing distinct feature categories. When both the spectral planes were scanned and their spatial features were enhanced via weighted fusion, it was observed that while the OA remained consistent compared to the dual-plane approach, the AA increased substantially with a 0.71 improvement over the H-W and H-C combinations. These results show that the tri-Mamba excels not only in modeling spatial features but also in effectively capturing spectral information. Thanks to its robust spectral feature modeling, the tri-Mamba achieved outstanding performance in detecting changes across all the categories.

D. Influence of T in SAM-KGFEM on Change Detection Results

To examine the effect of the threshold T in SAM-KGFEM on the overall experimental outcomes, a series of experiments were conducted using the China dataset with varying values of T. The results are shown in Table VIII, where T was preset to 0.5, 0.3, and 0.1 for comparison. While the OA results were similar across these threshold values, the AA results differed significantly. When T was set to 0.5 or 0.3, the threshold was too high, leading to an excessive fusion of masks as priors. This resulted in inaccurate prior masks, introduced noise, and hindered the enhancement of change characteristics in small-target areas. When T is set to 0.05, the threshold is too low, resulting in missing masks for the relevant bands of the small-sample category. This has led to the detection of small-sample classes being affected, resulting in a very low AA score. In contrast, when T was set to 0.1, the prior mask became more accurate with less noise, especially in detecting changes in small-target areas. This improvement in accuracy resulted in a higher AA score. These findings emphasize that selecting an appropriate threshold T is key to generating a suitable prior mask that enhances the detection of change areas.

TABLE IX

FREQUENCY SSM ABLATION RESULTS FROM THE CHINA DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUE HIGHLIGHTED IN RED

	w/o-SSM	High	Low
OA	98.29±0.18	98.18±0.15	98.28±0.03
AA	96.78±0.27	95.52±0.34	97.11±0.23
Kappa	96.13±0.14	95.96±0.16	96.18±0.06

TABLE X

PROMPT IN SAM ABLATION RESULTS FROM THE CHINA DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUE HIGHLIGHTED IN RED

	w/o-prompt	w-prompt
OA	98.22±0.05	98.28±0.03
AA	93.25±0.37	97.11±0.23
Kappa	96.06±0.13	96.18±0.06

E. Influence of SSM on Change Detection Results in Different Frequency Domains

The low-frequency domain contains relatively rich image information. The high-frequency domain contains limited texture information and significant noise [60], making it unsuitable for modeling with SSM. To further validate the effectiveness of the proposed approach, ablation studies were conducted on the WMAB module. As reported in Table IX, $w/o-$ SSM indicates that the SSM component is completely removed, while high and low represent the application of SSM in the high-frequency and low-frequency domains, respectively. Compared with the baseline without SSM, incorporating SSM in the high-frequency domain leads to a decline in OA, AA, and Kappa. This performance degradation is primarily attributed to the excessive noise inherent in high-frequency features, which adversely affects the stability and reliability of SSM modeling. In contrast, applying SSM in the low-frequency domain yields improvements in both AA and Kappa, with a marginal impact on OA. These results suggest that low-frequency features provide more stable and informative representations for SSM-based modeling.

F. Impact of Prompt in SAM on Change Detection Results

SAM is a general-purpose image segmentation framework capable of extracting target regions from a single image based on user-provided prompt. As illustrated in Table X, the integration of prompt significantly enhances performance in the context of change detection. Without prompt, SAM tends to produce inaccurate segmentation results, leading to a substantial number of misclassified regions. Consequently, a notable decline is observed in OA and AA. In contrast,

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the use of prompt effectively guides the model by explicitly indicating the target location, thereby reducing background interference and improving the precision of boundary delineation in change regions.

V. Conclusion

This article presents SFMS, a hyperspectral multiclass change detection method that addresses the issue of missed and false detections caused by insufficient training samples in multiclass classification problems. Unlike other Mamba-based methods, a gated Mamba with multiplane scanning is introduced, which emphasizes the importance of spectral information in HSI. The spatial state model is also combined with the frequency domain to extract complementary information from different frequency ranges. Furthermore, SAM is adapted for HSI segmentation. Given the spectral redundancy in HSIs, band selection is used to acquire more accurate prior knowledge to the change detection process. Both the qualitative and quantitative analyses are performed on three hyperspectral datasets to demonstrate the effectiveness of the proposed method. In conclusion, the proposed spatial–spectral feature–enhanced Mamba and SAM-guided hyperspectral multiclass change detection approach delivers strong results across all the test sets, validating the effectiveness of the tri-Mamba, WMAB, and SAM-KGFEM for improving multiclass change detection.

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